

Changes in KWN program - summary

July 4, 2024

The small modifications made in the code are gathered here, along with the parameters you need to provide

1 Modifications

1.1 Flow stress

Calculated in the program in the `src/KWN_model_routines.f90` file, lines 42 to 48.

1.1.1 Flow stress law

The flow stress σ can be calculated as:

$$\sigma = \sigma_0 + b\epsilon^{n_z} \quad (1)$$

where σ_0 , b , and n_z are input parameters, and ϵ is the strain

1.1.2 How to activate it

To activate it, you need to

- Make sure you provide `sigma_0`, `b` and `n_z` in the `namelist.input` file (please see the example in the `test_Ziyu` folder)
- Make sure you do **not** provide any parameter called `sigma_r` in the `namelist.input` (otherwise, the flow stress will be calculated as it was before with the sinepower law that depends on strain rate and temperature)

1.2 Dislocation density law

Calculated in the program in the `src/KWN_model_functions.f90` file, lines 25 to 48.

1.2.1 Dislocation density law

The empirically determined dislocation density law is given by

$$\rho = \frac{1.56 \cdot 10^{14}}{1 + \exp(-19.1(\epsilon - 0.24))} \cdot \frac{1}{M_z^2} \quad (2)$$

where the only input parameter here is M_z , and the rest is hard-coded (that can be changed if needed, let me know if you want some of these parameters to be adjustable)

1.2.2 How to activate it

To activate it, you need to

- Add the parameter `empirical_law_dislocation_density=1` in the `namelist.input` file.
- Provide the `M_z` parameter in the `namelist.input` file (default value is 1)

1.3 Pipe diffusion

1.3.1 Diffusion coefficient law

In addition to the contribution of excess vacancies, a supplementary term D_p is added to the diffusion coefficient that is calculated as :

$$D_d = 2 \cdot \frac{\rho \Omega}{b} \cdot D_0 \exp -Q_d/RT \quad (3)$$

where ρ is the dislocation density, D_0 is the same as for bulk diffusion. For the parameters in the `namelist.input` file

- Ω is the atomic volume of the matrix (`atomic_volume`, [m³])
- b is the Burgers vector (`burgers` [m])
- Q_d is the activation energy for solute migration around dislocation core (`q_dislocation` [J/mol])

1.3.2 How to activate it

It is activated by default. If you want to get rid of it, you can set Q_d (`q_dislocation`) to a very high value.